SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION SYSTEMS

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FAT

AGILE DEVELOPMENT PROCESS AND DEVOPS LAB

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SET-6

Smart Electricity Billing System

Problem statement:

Ensure electricity usage is calculated and billed correctly

Tasks:

1.Develop billing calculation logic

2. Write Junit tests for tariff calculations

3.Implement CI/CD pipeline with automation testing

4.Deploy services on Docker and Kubernetes

App.java

```
package com.example;


public class App {
   public static final int SLAB1_LIMIT = 100;
   public static final int SLAB2_LIMIT = 300;
   public static final int SLAB3_LIMIT = 500;
   public static final double RATE_SLAB1 = 1.50;
   public static final double RATE_SLAB2 = 2.50;
   public static final double RATE_SLAB3 = 4.00;
   public static final double RATE_SLAB4 = 6.00;
   public static final double FIXED_METER_CHARGE = 50.00;
   public static final double TAX_RATE = 0.05; // 5%


   public static int calculateUnitsConsumed(int currentReading, int previousReading) {
      if (currentReading < 0 || previousReading < 0) {
         throw new IllegalArgumentException("Meter readings cannot be negative.");
      }
      if (currentReading < previousReading) {
         throw new IllegalArgumentException("Current reading cannot be less than
previous reading.");
      }
      return currentReading - previousReading;
   }
   public static double calculateEnergyCharge(int units) {
      if (units < 0) {
         throw new IllegalArgumentException("Units consumed cannot be negative.");
      }
```

```
    double charge = 0.0;
    int remaining = units;
    if (remaining > 0) {
        int slabUnits = Math.min(remaining, SLAB1_LIMIT);
        charge += slabUnits * RATE_SLAB1;
        remaining -= slabUnits;
    }
    if (remaining > 0) {
        int slabUnits = Math.min(remaining, SLAB2_LIMIT - SLAB1_LIMIT);
        charge += slabUnits * RATE_SLAB2;
        remaining -= slabUnits;
    }
    if (remaining > 0) {
        int slabUnits = Math.min(remaining, SLAB3_LIMIT - SLAB2_LIMIT);
        charge += slabUnits * RATE_SLAB3;
        remaining -= slabUnits;
    }
    if (remaining > 0) {
        charge += remaining * RATE_SLAB4;
    }

    return charge;
}

    public static double calculateTax(double energyCharge) {
    return energyCharge * TAX_RATE;
}
```

```
    public static double calculateTotalBill(int units) {
        double energyCharge = calculateEnergyCharge(units);
        double tax = calculateTax(energyCharge);
        return energyCharge + tax + FIXED_METER_CHARGE;
    }
    public static String generateBill(String customerName, int currentReading, int
previousReading) {
        int units = calculateUnitsConsumed(currentReading, previousReading);
        double energyCharge = calculateEnergyCharge(units);
        double tax = calculateTax(energyCharge);
        double totalBill = calculateTotalBill(units);

        StringBuilder sb = new StringBuilder();
        sb.append("========================================\n");
        sb.append("      ELECTRICITY BILL SUMMARY       \n");
        sb.append("========================================\n");
        sb.append(String.format("Customer Name    : %s%n", customerName));
        sb.append(String.format("Previous Reading : %d kWh%n", previousReading));
        sb.append(String.format("Current Reading  : %d kWh%n", currentReading));
        sb.append(String.format("Units Consumed   : %d units%n", units));
        sb.append("----------------------------------------\n");
        sb.append(String.format("Energy Charge    : ₹%.2f%n", energyCharge));
        sb.append(String.format("Tax (5%%)        : ₹%.2f%n", tax));
        sb.append(String.format("Meter Charge     : ₹%.2f%n", FIXED_METER_CHARGE));
        sb.append("----------------------------------------\n");
        sb.append(String.format("TOTAL AMOUNT DUE : ₹%.2f%n", totalBill));
        sb.append("========================================\n");
        return sb.toString();
```

```
    }

    public static void main(String[] args) {
        System.out.println("Electricity Billing System initialized.\n");

        // Example bill generation
        String bill = generateBill("Naveen Kumar", 5350, 4900);
        System.out.println(bill);
    }
}
```

AppTest.java:

```
package com.example;

import junit.framework.TestCase;
public class AppTest extends TestCase {
    public void testCalculateUnitsConsumed_NormalUsage() {
        assertEquals(450, App.calculateUnitsConsumed(5350, 4900));
    }

    public void testCalculateUnitsConsumed_ZeroUsage() {
        assertEquals(0, App.calculateUnitsConsumed(1000, 1000));
    }

    public void testCalculateUnitsConsumed_SmallUsage() {
        assertEquals(1, App.calculateUnitsConsumed(101, 100));
    }
```

```
public void testCalculateUnitsConsumed_LargeUsage() {
    assertEquals(10000, App.calculateUnitsConsumed(20000, 10000));
}

public void testCalculateUnitsConsumed_NegativeCurrentReading() {
    try {
        App.calculateUnitsConsumed(-100, 50);
        fail("Expected IllegalArgumentException for negative current reading");
    } catch (IllegalArgumentException e) {
        assertEquals("Meter readings cannot be negative.", e.getMessage());
    }
}

public void testCalculateUnitsConsumed_NegativePreviousReading() {
    try {
        App.calculateUnitsConsumed(100, -50);
        fail("Expected IllegalArgumentException for negative previous reading");
    } catch (IllegalArgumentException e) {
        assertEquals("Meter readings cannot be negative.", e.getMessage());
    }
}

public void testCalculateUnitsConsumed_CurrentLessThanPrevious() {
    try {
        App.calculateUnitsConsumed(100, 200);
        fail("Expected IllegalArgumentException when current < previous");
    } catch (IllegalArgumentException e) {
```

```
        assertEquals("Current reading cannot be less than previous reading.",
e.getMessage());
    }
  }

  public void testEnergyCharge_ZeroUnits() {
    assertEquals(0.0, App.calculateEnergyCharge(0), 0.01);
  }

  public void testEnergyCharge_Slab1_OneUnit() {
    assertEquals(1.50, App.calculateEnergyCharge(1), 0.01);
  }

  public void testEnergyCharge_Slab1_FiftyUnits() {
    assertEquals(75.00, App.calculateEnergyCharge(50), 0.01);
  }

  public void testEnergyCharge_Slab1_BoundaryExact() {
    assertEquals(150.00, App.calculateEnergyCharge(100), 0.01);
  }

  public void testEnergyCharge_Slab2_BoundaryStart() {
    assertEquals(152.50, App.calculateEnergyCharge(101), 0.01);
  }

  public void testEnergyCharge_Slab2_MidRange() {
    assertEquals(400.00, App.calculateEnergyCharge(200), 0.01);
  }
```

```
public void testEnergyCharge_Slab2_BoundaryExact(){
   assertEquals(650.00, App.calculateEnergyCharge(300), 0.01);
}
public void testEnergyCharge_Slab3_BoundaryStart() {
   assertEquals(654.00, App.calculateEnergyCharge(301), 0.01);
}

public void testEnergyCharge_Slab3_MidRange() {
   assertEquals(1050.00, App.calculateEnergyCharge(400), 0.01);
}

public void testEnergyCharge_Slab3_450Units() {
   assertEquals(1250.00, App.calculateEnergyCharge(450), 0.01);
}

public void testEnergyCharge_Slab3_BoundaryExact() {
   assertEquals(1450.00, App.calculateEnergyCharge(500), 0.01);
}
public void testEnergyCharge_Slab4_BoundaryStart() {
   assertEquals(1456.00, App.calculateEnergyCharge(501), 0.01);
}

public void testEnergyCharge_Slab4_LargeUsage() {
   // 100 × 1.50 + 200 × 2.50 + 200 × 4.00 + 500 × 6.00
   // = 150 + 500 + 800 + 3000 = 4450.00
   assertEquals(4450.00, App.calculateEnergyCharge(1000), 0.01);
}
```

```
public void testEnergyCharge_NegativeUnits() {
    try {
        App.calculateEnergyCharge(-10);
        fail("Expected IllegalArgumentException for negative units");
    } catch (IllegalArgumentException e) {
        assertEquals("Units consumed cannot be negative.", e.getMessage());
    }
}

public void testTax_ZeroCharge() {
    assertEquals(0.0, App.calculateTax(0.0), 0.01);
}

public void testTax_Slab1() {
    // Energy charge for 100 units = 150.00, tax = 150 × 0.05 = 7.50
    double energyCharge = App.calculateEnergyCharge(100);
    assertEquals(7.50, App.calculateTax(energyCharge), 0.01);
}

public void testTax_Slab3() {
    // Energy charge for 450 units = 1250.00, tax = 1250 × 0.05 = 62.50
    double energyCharge = App.calculateEnergyCharge(450);
    assertEquals(62.50, App.calculateTax(energyCharge), 0.01);
}

public void testTax_LargeCharge() {
    // Energy charge for 1000 units = 4450.00, tax = 4450 × 0.05 = 222.50
```

```
    double energyCharge = App.calculateEnergyCharge(1000);
    assertEquals(222.50, App.calculateTax(energyCharge), 0.01);
}
public void testTotalBill_ZeroUnits() {
    assertEquals(50.00, App.calculateTotalBill(0), 0.01);
}

public void testTotalBill_Slab1Only() {
    assertEquals(207.50, App.calculateTotalBill(100), 0.01);
}

public void testTotalBill_Slab2Boundary() {
    assertEquals(732.50, App.calculateTotalBill(300), 0.01);
}

public void testTotalBill_Slab3_450Units() {
    assertEquals(1362.50, App.calculateTotalBill(450), 0.01);
}

public void testTotalBill_Slab4_1000Units() {
    assertEquals(4722.50, App.calculateTotalBill(1000), 0.01);
}
public void testGenerateBill_ContainsCustomerName() {
    String bill = App.generateBill("Naveen Kumar", 5350, 4900);
    assertTrue(bill.contains("Naveen Kumar"));
}

public void testGenerateBill_ContainsUnitsConsumed() {
```

```
        String bill = App.generateBill("Test User", 1200, 1000);
        assertTrue(bill.contains("200 units"));
    }

    public void testGenerateBill_ContainsTotalAmount() {
        String bill = App.generateBill("Test User", 1100, 1000);
        // 100 units → Energy = 150, Tax = 7.50, Meter = 50 → Total = 207.50
        assertTrue(bill.contains("207.50"));
    }

    public void testGenerateBill_ContainsMeterReadings() {
        String bill = App.generateBill("Test User", 5000, 4500);
        assertTrue(bill.contains("5000"));
        assertTrue(bill.contains("4500"));
    }
}
```

POM.XML:

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>electricity-billing</artifactId>
  <packaging>jar</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>electricity-billing</name>
```

```
<url>http://maven.apache.org</url>

<properties>
  <maven.compiler.source>17</maven.compiler.source>
  <maven.compiler.target>17</maven.compiler.target>
  <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
</properties>

<dependencies>
  <dependency>
    <groupId>junit</groupId>
    <artifactId>junit</artifactId>
    <version>4.13.2</version>
    <scope>test</scope>
  </dependency>
</dependencies>

<build>
  <plugins>
    <!-- Compiler plugin -->
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-compiler-plugin</artifactId>
      <version>3.11.0</version>
      <configuration>
        <source>17</source>
        <target>17</target>
      </configuration>
```

```
    </plugin>

    <!-- Surefire plugin for running tests -->
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-surefire-plugin</artifactId>
      <version>3.1.2</version>
    </plugin>

    <!-- JAR plugin with main class manifest -->
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-jar-plugin</artifactId>
      <version>3.3.0</version>
      <configuration>
        <archive>
          <manifest>
            <mainClass>com.example.App</mainClass>
          </manifest>
        </archive>
      </configuration>
    </plugin>
  </plugins>
</build>
</project>
```

Junit Test:

```
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 34, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.112 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 34, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.646 s
[INFO] Finished at: 2026-04-15T18:01:31+05:30
[INFO] ------------------------------------------------------------------------
PS D:\Study_Material\SEM-6\Agile\lab\FAT\Multi_Factor_Authentication-main>
```

Local Build:

```
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 34, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.084 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 34, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.3.0:jar (default-jar) @ electricity-billing ---
[INFO] Building jar: D:\Study_Material\SEM-6\Agile\lab\FAT\Multi_Factor_Authentication-main\target\electricity-billing-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.641 s
[INFO] Finished at: 2026-04-15T18:00:45+05:30
[INFO] ------------------------------------------------------------------------
PS D:\Study_Material\SEM-6\Agile\lab\FAT\Multi_Factor_Authentication-main>
```

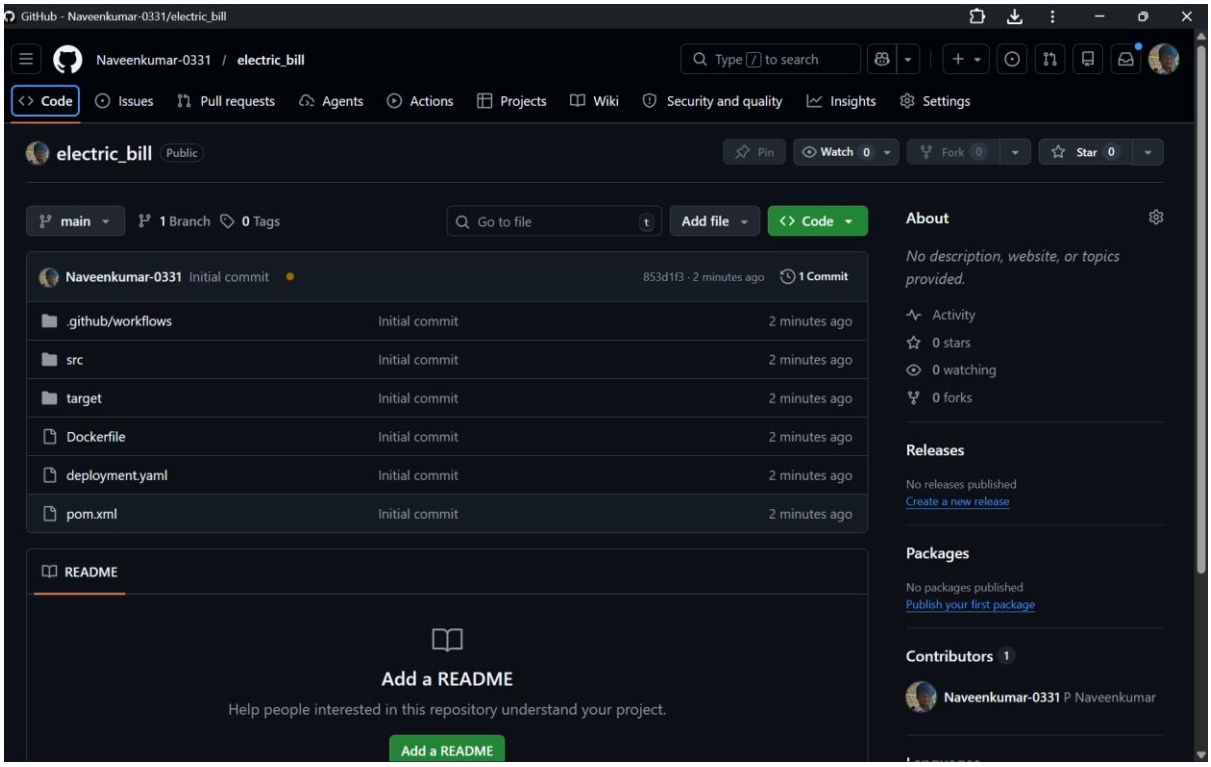
Git repo link : https://github.com/Naveenkumar-0331/electric_bill.git

GitHub init and commit:

```
PS D:\Study_Material\SEM-6\Agile\lab\FAT\electric_bill> git init
Initialized empty Git repository in D:/Study_Material/SEM-6/Agile/lab/FAT/electric_bill/.git/
PS D:\Study_Material\SEM-6\Agile\lab\FAT\electric_bill> git add .
warning: in the working copy of '.github/workflows/ci-cd.yml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'Dockerfile', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'deployment.yaml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'pom.xml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/example/App.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/test/java/com/example/AppTest.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst', LF will be replaced by CRLF the next time Git touches
 it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst', LF will be replaced by CRLF the next time Git touches i
t
warning: in the working copy of 'target/surefire-reports/TEST-com.example.AppTest.xml', LF will be replaced by CRLF the next time Git touches it
PS D:\Study_Material\SEM-6\Agile\lab\FAT\electric_bill> git commit -m "Initial commit"
[master (root-commit) 853d1f3] Initial commit
 18 files changed, 743 insertions(+)
 create mode 100644 .github/workflows/ci-cd.yml
 create mode 100644 Dockerfile
 create mode 100644 deployment.yaml
 create mode 100644 pom.xml
 create mode 100644 src/main/java/com/example/App.java
 create mode 100644 src/test/java/com/example/AppTest.java
 create mode 100644 target/classes/com/example/App.class
 create mode 100644 target/electricity-billing-1.0-SNAPSHOT.jar
 create mode 100644 target/maven-archiver/pom.properties
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
```

```
PS D:\Study_Material\SEM-6\Agile\lab\FAT\electric_bill> git commit -m "Initial commit"
 create mode 100644 target/surefire-reports/TEST-com.example.AppTest.xml
 create mode 100644 target/surefire-reports/com.example.AppTest.txt
 create mode 100644 target/test-classes/com/example/AppTest.class
PS D:\Study_Material\SEM-6\Agile\lab\FAT\electric_bill> git branch -M main
PS D:\Study_Material\SEM-6\Agile\lab\FAT\electric_bill> git remote add origin https://github.com/Naveenkumar-0331/electric_bill.git
PS D:\Study_Material\SEM-6\Agile\lab\FAT\electric_bill> git push -u origin main
Enumerating objects: 46, done.
Counting objects: 100% (46/46), done.
Delta compression using up to 18 threads
Compressing objects: 100% (24/24), done.
Writing objects: 100% (46/46), 16.57 KiB | 628.00 KiB/s, done.
Total 46 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To https://github.com/Naveenkumar-0331/electric_bill.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS D:\Study_Material\SEM-6\Agile\lab\FAT\electric_bill>
```

Github repo:



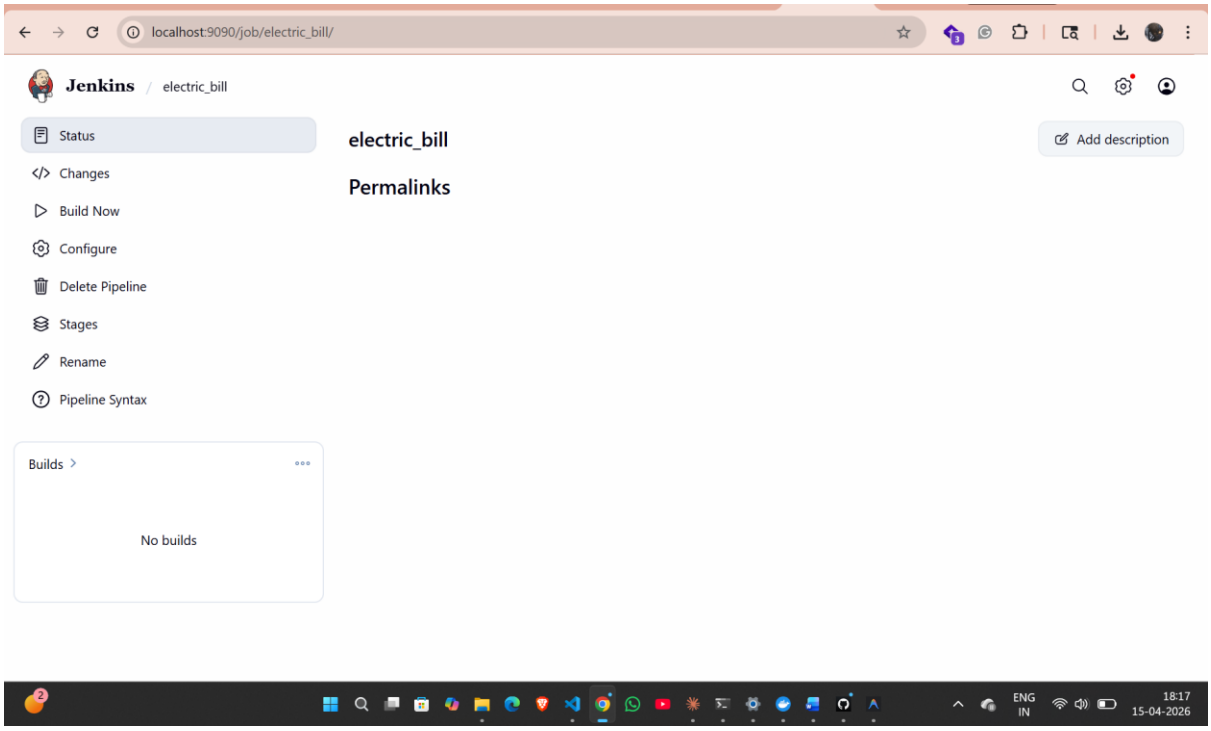
Jenkins:

Pipeline code:

```
pipeline {
 agent any


 environment {
 DOCKER_IMAGE = "naveen14k/java-test-app:latest"
 KUBECONFIG = "C:\\ProgramData\\Jenkins\\.kube\\config"
 }
 stages {
 stage('Clone Repository') {
 steps {
```

```
git branch: 'main',
url: 'https://github.com/Naveenkumar-0331/electric_bill.git'
}
}
stage('Build & Run Tests') {
steps {
bat 'mvn clean test'
}
post {
always {
junit 'target/surefire-reports/*.xml'
}
}
}
stage('Package Application') {
steps {
bat 'mvn package -DskipTests'
}
}
stage('Build Docker Image') {
steps {
bat "docker build -t %DOCKER_IMAGE% ."
}
}
stage('Push to DockerHub') {
steps {
withCredentials([usernamePassword(
credentialsId: 'docker',
```

```
usernameVariable: 'DOCKER_USER',
passwordVariable: 'DOCKER_PASS'
)]) {
bat """
docker login -u %DOCKER_USER% -p %DOCKER_PASS%
docker push %DOCKER_IMAGE%
"""
}
}
}
stage('Deploy to Kubernetes') {
steps {
bat 'kubectl apply -f deployment.yaml --validate=false'
}
}
}
}
```

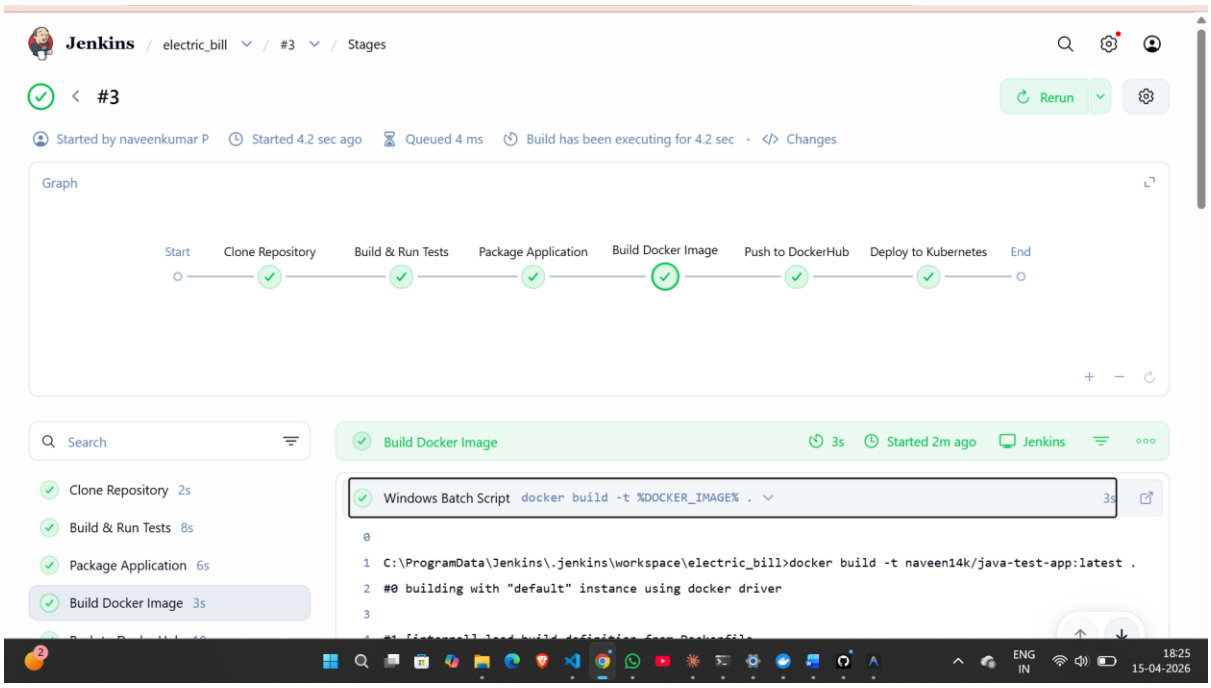


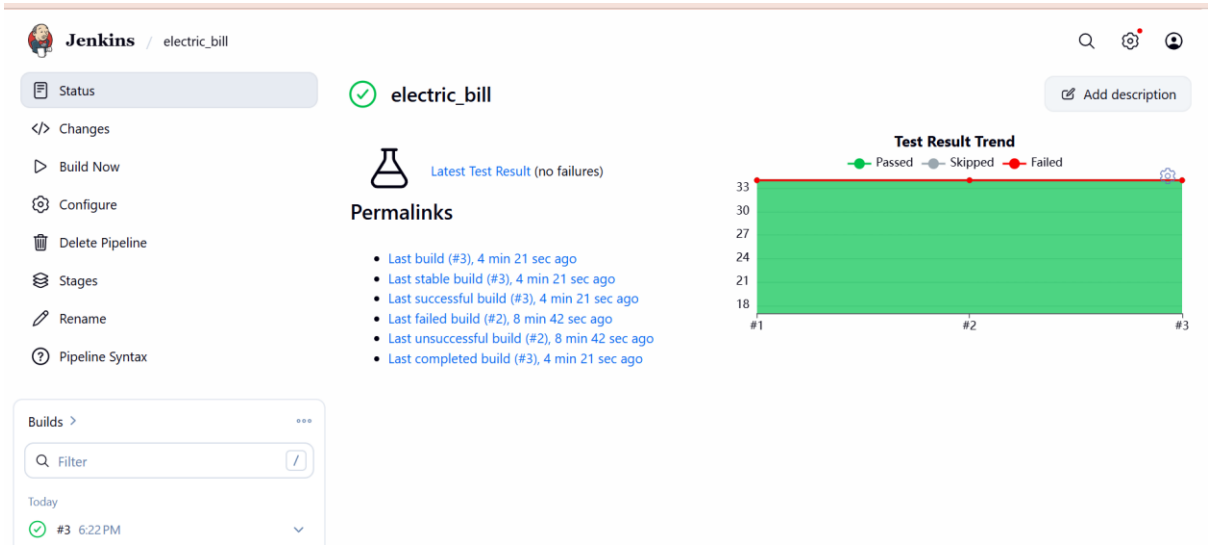
Maven Build in Jenkin:

Automation Test in Jenkin:

```
148 [INFO]
149 [INFO] -------------------------------------------------------
150 [INFO]  T E S T S
151 [INFO] -------------------------------------------------------
152 [INFO] Running com.example.AppTest
153 [INFO] Tests run: 34, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.166 s -- in com.example.AppTest
154 [INFO]
155 [INFO] Results:
156 [INFO]
157 [INFO] Tests run: 34, Failures: 0, Errors: 0, Skipped: 0
158 [INFO]
```

CI/CD pipeline in jenkin:

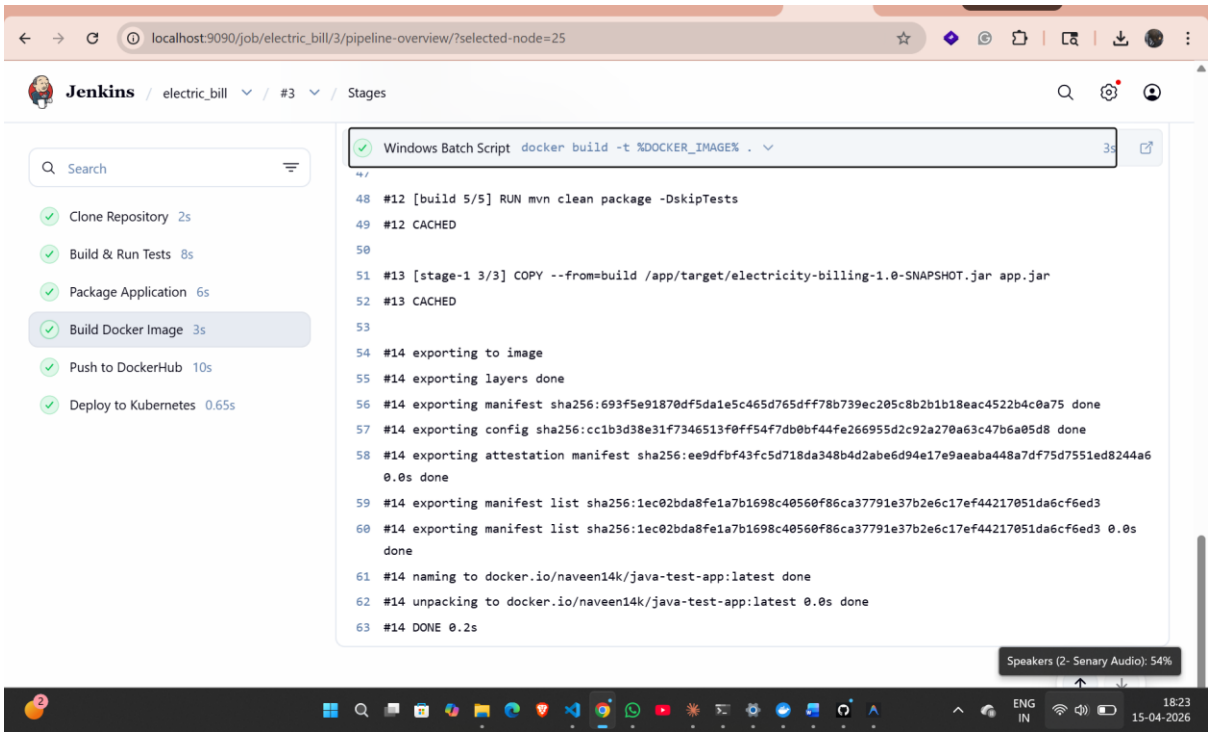




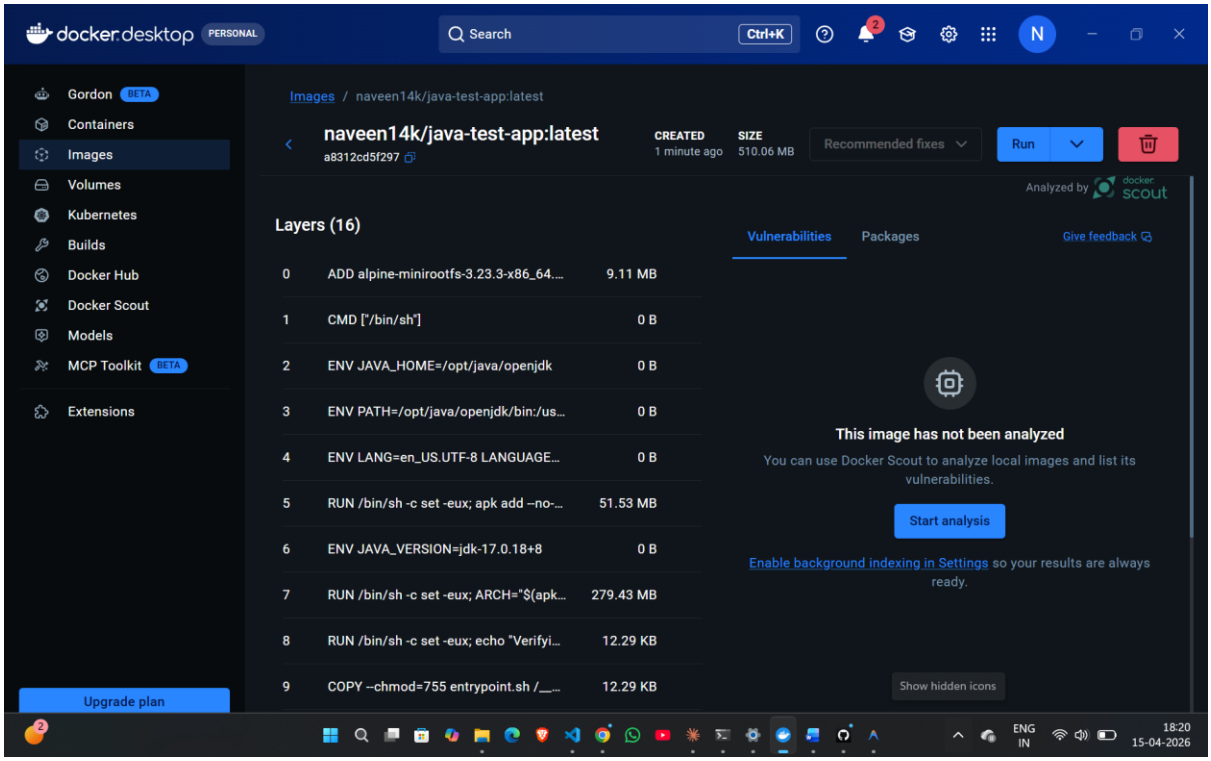
Dockerfile:

```
# Stage 1: Build the application

FROM maven:3.9-eclipse-temurin-17-alpine AS build

WORKDIR /app

COPY pom.xml .

COPY src ./src

RUN mvn clean package -DskipTests


# Stage 2: Run the application

FROM eclipse-temurin:17-jdk-alpine

WORKDIR /app

COPY --from=build /app/target/electricity-billing-1.0-SNAPSHOT.jar app.jar

EXPOSE 8080

ENTRYPOINT ["java", "-jar", "app.jar"]
```

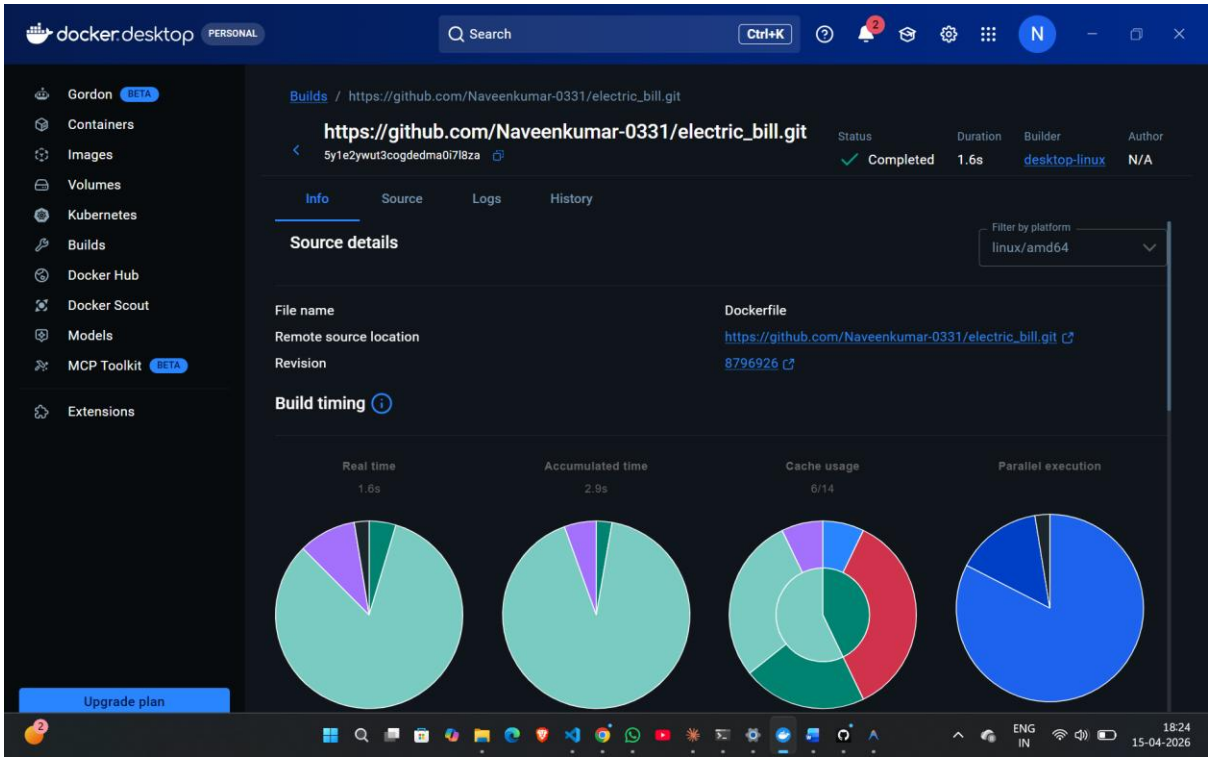
Image creation in jenkin:



Docker Hub Image:



Docker Build:



Docker Local Build:

```
PS D:\Study_Material\SEM-6\Agile\lab\FAT\electric_bill> docker build -t naveen14k/electricity-billing:latest .
>>
 => => transferring dockerfile: 405B                                                                                    0.0s
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk-alpine                                       3.3s
 => [internal] load metadata for docker.io/library/maven:3.9-eclipse-temurin-17-alpine                                 3.3s
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.io                                                 0.0s
 => [auth] library/maven:pull token for registry-1.docker.io                                                            0.0s
 => [internal] load .dockerignore                                                                                       0.1s
 => => transferring context: 2B                                                                                         0.0s
 => [stage-1 1/3] FROM docker.io/library/eclipse-temurin:17-jdk-alpine@sha256:903e865ecd125a01759fd156d042b998aa86957262f6dec47451  0.0s
 => => resolve docker.io/library/eclipse-temurin:17-jdk-alpine@sha256:903e865ecd125a01759fd156d042b998aa86957262f6dec4745125a47bd4  0.0s
 => [internal] load build context                                                                                       0.0s
 => => transferring context: 16.76kB                                                                                    0.0s
 => [build 1/5] FROM docker.io/library/maven:3.9-eclipse-temurin-17-alpine@sha256:f76c19900a1809ef6d1e13cc7e9dd30c4a5be854f4abd87f  0.0s
 => => resolve docker.io/library/maven:3.9-eclipse-temurin-17-alpine@sha256:f76c19900a1809ef6d1e13cc7e9dd30c4a5be854f4abd87faacd67  0.0s
 => CACHED [build 2/5] WORKDIR /app                                                                                     0.0s
 => [build 3/5] COPY pom.xml .                                                                                          0.0s
 => [build 4/5] COPY src ./src                                                                                          0.1s
 => [build 5/5] RUN mvn clean package -DskipTests                                                                      30.9s
 => CACHED [stage-1 2/3] WORKDIR /app                                                                                   0.0s
 => [stage-1 3/3] COPY --from=build /app/target/electricity-billing-1.0-SNAPSHOT.jar app.jar                            0.1s
 => exporting to image                                                                                                  0.3s
 => => exporting layers                                                                                                 0.1s
 => => exporting manifest sha256:cc784b86f5e154278dab785382e7fdf4434ad83b9f2c155f35e84d47efa55d63                       0.0s
 => => exporting config sha256:7bd34a368a0cedd88fa604964d2c359f8ac0a099433ee435866fcd3f3bbe5ecc                         0.0s
 => => exporting attestation manifest sha256:0dddbee041c4d58b9c6156f46a1de8f6306ed30b87270f10252f3419dc08f380           0.0s
 => => exporting manifest list sha256:9017f63844cb51bfd52d2837982951a6948a864e9e5dc224a6c7cc715517fefb                  0.0s
 => => naming to docker.io/naveen14k/electricity-billing:latest                                                         0.0s
 => => unpacking to docker.io/naveen14k/electricity-billing:latest                                                      0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/znpref6mm81flzbf1l808g45t
PS D:\Study_Material\SEM-6\Agile\lab\FAT\electric_bill>
```

Kubernetes

Deployment.yaml:

apiVersion: apps/v1

kind: Deployment

metadata:

  name: electricity-billing-deployment

  labels:

    app: electricity-billing

spec:

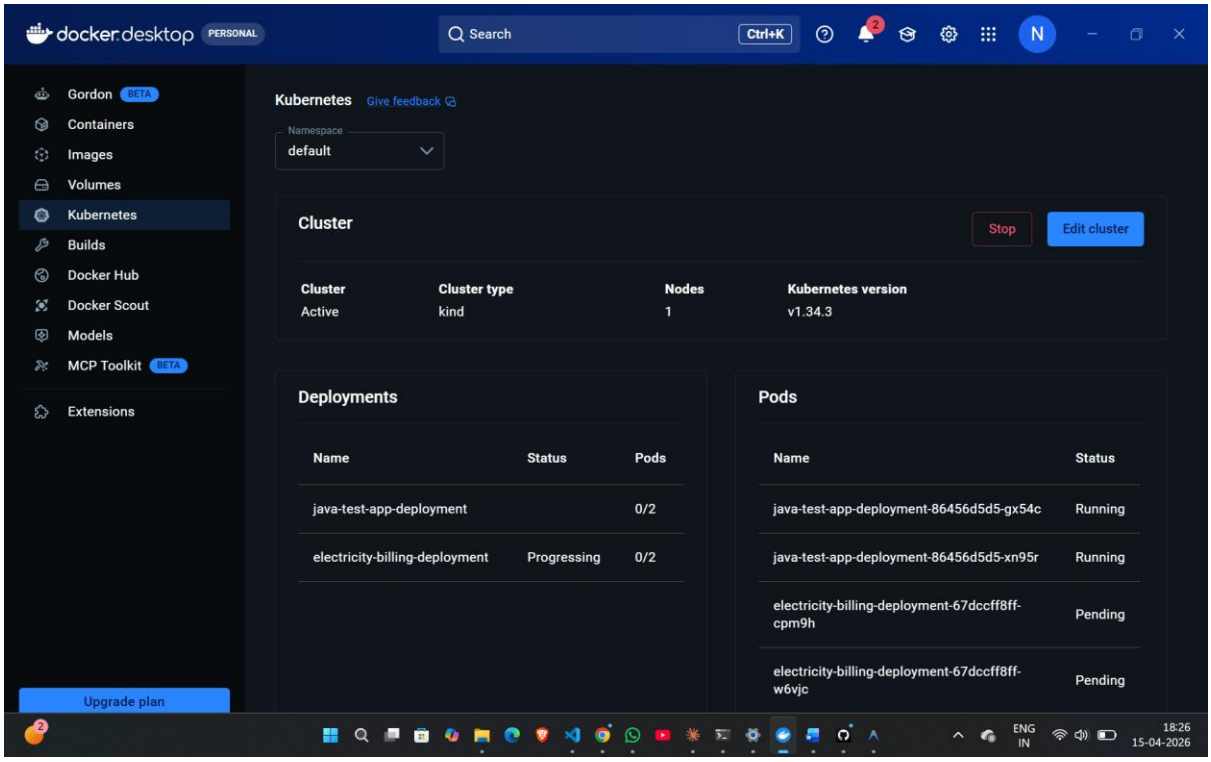
  replicas: 2

  selector:

```
  matchLabels:
    app: electricity-billing

  template:
    metadata:
      labels:
        app: electricity-billing

    spec:
      containers:
        - name: electricity-billing
          image: naveen14k/electricity-billing:latest
          imagePullPolicy: Always

          ports:
            - containerPort: 8080

          resources:
            requests:
              memory: "256Mi"
              cpu: "250m"
            limits:
              memory: "512Mi"
              cpu: "500m"

          livenessProbe:
            tcpSocket:
              port: 8080
```

```
          initialDelaySeconds: 30
          periodSeconds: 10

        readinessProbe:
          tcpSocket:
            port: 8080
          initialDelaySeconds: 10
          periodSeconds: 5

---
apiVersion: v1
kind: Service
metadata:
  name: electricity-billing-service

spec:
  type: NodePort
  selector:
    app: electricity-billing

  ports:
    - protocol: TCP
      port: 80
      targetPort: 8080
      nodePort: 30009
```

Kubernetes:



Kubernets Local:

```
PS D:\Study_Material\SEM-6\Agile\lab\FAT\electric_bill> kubectl apply -f deployment.yaml
>>
deployment.apps/electricity-billing-deployment unchanged
service/electricity-billing-service unchanged
PS D:\Study_Material\SEM-6\Agile\lab\FAT\electric_bill> # Check if your pods are running:
>> kubectl get pods
>>
>> # Check if the service was created and see the exposed port:
>> kubectl get svc electricity-billing-service
>>
NAME                                          READY   STATUS             RESTARTS        AGE
electricity-billing-deployment-67dccff8ff-cpm9h   0/1     ImagePullBackOff   0               11m
electricity-billing-deployment-67dccff8ff-w6vjc   0/1     ImagePullBackOff   0               11m
java-test-app-deployment-86456d5d5-gx54c          0/1     CrashLoopBackOff   11 (2m27s ago)   34m
java-test-app-deployment-86456d5d5-xn95r          0/1     CrashLoopBackOff   11 (2m40s ago)   34m
NAME                          TYPE       CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
electricity-billing-service   NodePort   10.96.195.241   <none>        80:30009/TCP   7m27s
```